

Question number	Answer	Additional guidance	Mark
2(a)	0.989 / 0.99 / 1.0 / 1 (m)	ACCEPT 0.994	(1)

Question number	Answer	Additional guidance	Mark										
2(b)	<table><tr><th>Hazard</th><th>How to reduce the risk</th></tr><tr><td>(using / cutting oneself with) {knife / scissors / scalpel}</td><td>cut {away from fingers / hands / oneself / on a board or tile / wearing gloves} (1)</td></tr><tr><td>OR</td><td></td></tr><tr><td>(contact with / allergy to) fish (1)</td><td>wear {gloves / mask}</td></tr><tr><td>(use of / inhalation of / spilling of) formalin (1)</td><td>use in a well-ventilated room / use in a fume cupboard / wear gloves / wash hands / wear protective eye wear / wear mask (1)</td></tr></table>	Hazard	How to reduce the risk	(using / cutting oneself with) {knife / scissors / scalpel}	cut {away from fingers / hands / oneself / on a board or tile / wearing gloves} (1)	OR		(contact with / allergy to) fish (1)	wear {gloves / mask}	(use of / inhalation of / spilling of) formalin (1)	use in a well-ventilated room / use in a fume cupboard / wear gloves / wash hands / wear protective eye wear / wear mask (1)	ACCEPT preservative liquid for formalin	(4)
Hazard	How to reduce the risk												
(using / cutting oneself with) {knife / scissors / scalpel}	cut {away from fingers / hands / oneself / on a board or tile / wearing gloves} (1)												
OR													
(contact with / allergy to) fish (1)	wear {gloves / mask}												
(use of / inhalation of / spilling of) formalin (1)	use in a well-ventilated room / use in a fume cupboard / wear gloves / wash hands / wear protective eye wear / wear mask (1)												

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2(c)	An answer that includes the following points to include 1 similarity and 1 difference: SIMILARITIES: • both (hearts) have atrium(a) and ventricle(s) (1) • (in both) ventricle wall(s) thicker (than atrium wall(s)) (1) • both have valves (1) DIFFERENCES: • fish heart has one {atrium / ventricle} whereas a mammalian heart has two {atria / ventricles} / fish has only one side to the heart, mammals have two / mammalian heart has septum, fish does not (1) • fish heart has a {sinus venosus / bulbus arteriosus}, but a mammalian heart does not (1)	Must be a clear statement ACCEPT Reverse argument ACCEPT fish has single, mammal has double circulation / fish has 2 chambers, mammal has 4 ACCEPT ...but mammals have arteries and veins	(3)

Question number	Answer	Additional guidance	Mark
2(d)	An answer that includes 6 of the following points: • use minimum of five fish (1) • (of) different sizes (1) • measure the {length / mass / volume} of each fish (1) • measure the {length / mass/ volume} of each heart (1) • detail of fish standardisation of measurement (1) • detail of standardisation of heart {measurement / dissection} (1) • plot a graph of fish size vs heart size / calculate ratios (1) • carry out a (correctly named) correlation test / compare ratios (1)	ACCEPT more than one quoted size, minimum a big one and a small one ACCEPT diameter ACCEPT diameter e.g., between same points on each fish, preparation of fish prior to measurement (e.g., remove parasites) e.g., length from end of sinus venosus to start of bulbus arteriosus each time, preparation of heart prior to measurement (e.g. empty of blood, remove extraneous tissues)	(6)